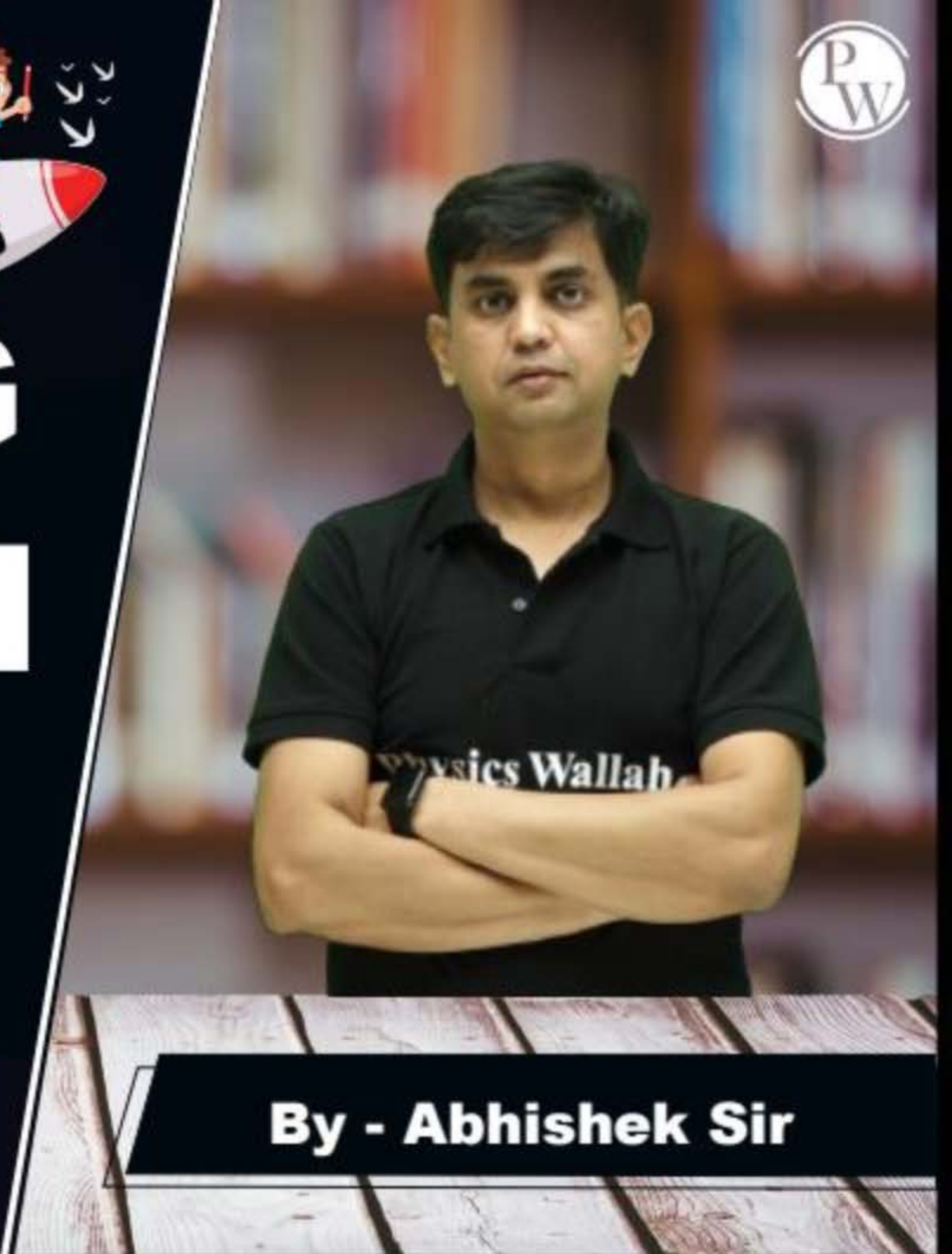
CS & IT ENGINEERING

Computer Network

Application Layer

Lecture No. - 03



By - Abhishek Sir

Recap of Previous Lecture

Topic DNS

Topic HTTP

Topics to be Covered

Topic HTTP

ABOUT ME

Hello, I'm Abhishek

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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Topic : HTTP Connections

→ Two types of HTTP connections :

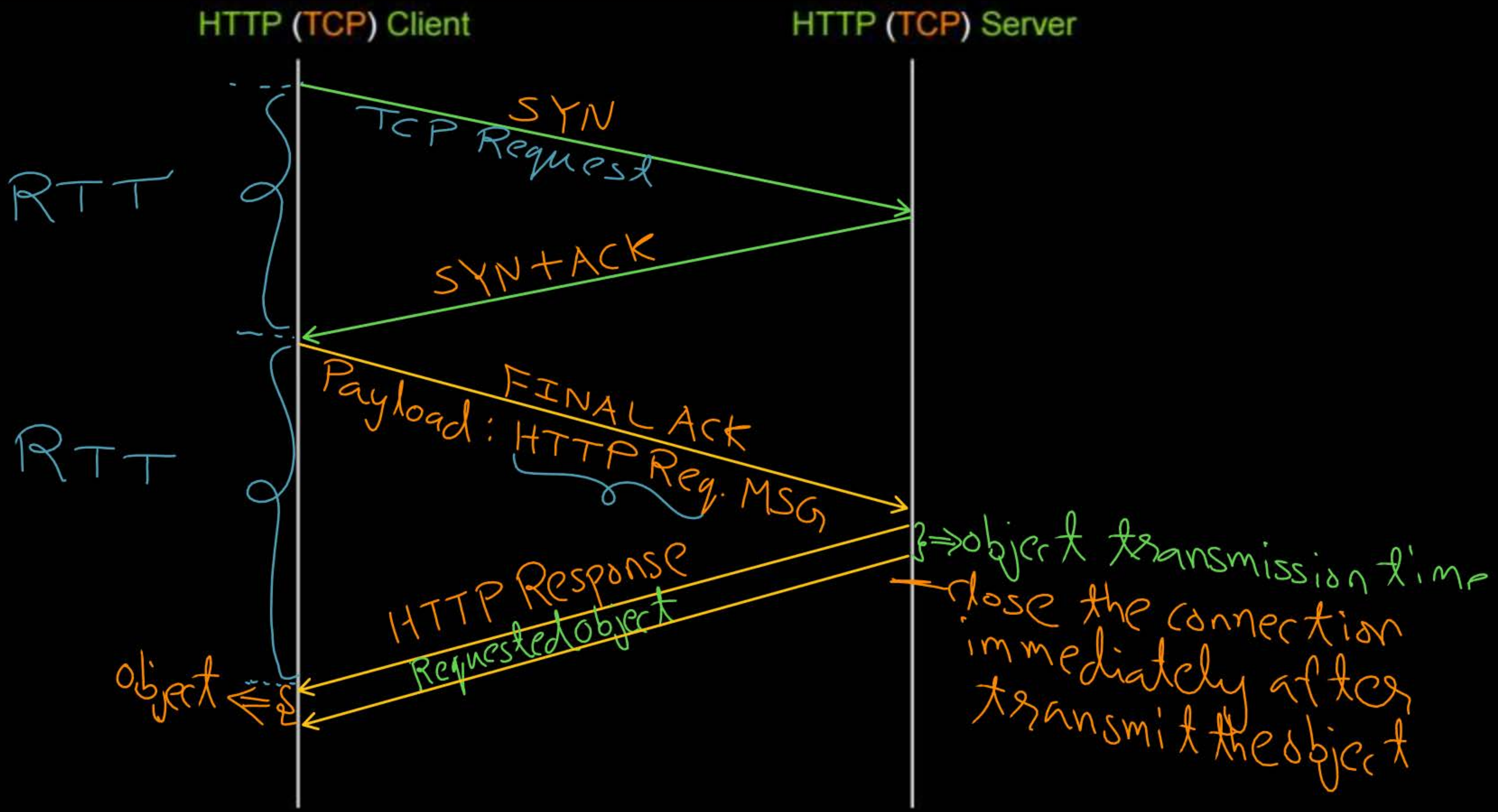
1. Non-persistent HTTP : $\boxed{\text{HTTP/1.0}}$
2. Persistent HTTP : HTTP/1.1

Topic : Non-persistent HTTP (HTTP/1.0)

→ At most one object sent over a TCP connection

→ Downloading multiple objects required multiple TCP connections

Topic : Non-persistent HTTP (HTTP/1.0)



Example 3 :-

#Q. Consider a web browser at client machine made a request for a web page to a web server using non-persistent HTTP. The web page contains base HTML file and ten very small image objects. How many TCP connections required to load the web page completely ?

Ans = $(1+10) = 11$ TCP Connections.

[NAT] $\boxed{\text{H.W.}}$ IIT-D [GATE-2020][1 Mark]

#Q. Assume that you have made a request for a web page through your web browser to a web server. Initially the browser cache is empty. Further, the browser is configured to send HTTP requests in non-persistent mode. The web page contains text and five very small images. The minimum number of TCP connections required to display the web page completely in your browser is ______.

Topic : Non-persistent HTTP (HTTP/1.0)

HTTP response time (per object) :

→ One RTT to initiate TCP connection

→ One RTT for HTTP request and first few bytes of HTTP response to return

→ Object / File transmission time

$$\boxed{\text{Response time} = [2 * \text{RTT}] + [\text{File Transmission Time}]}$$

Example 4 :-

#Q. Consider a web browser at client machine made a request for a web page to a web server using non-persistent HTTP (no any parallel TCP connections). The web page contains base HTML file and ten very small image objects. How many total number of RTTs (Round Trip Time) it takes to load the web page completely ?

① For Base HTML(Text) File :

1 RTT for TCP + 1 RTT for HTTP Req. & Response = 2 RTT

Base HTML file Have 10 small size image object reference

i) first image object → 1 RTT + 1 RTT = 2 RTT

ii) second image object → 1 RTT + 1 RTT = 2 RTT

⋮

x) Tenth image object → 1 RTT + 1 RTT = 2 RTT

Ans = (2 + 10 * 2) RTT = 22 RTT

Topic : Non-persistent HTTP (HTTP/1.0)

→ Requires 2 RTT per object

→ Operating System overhead for each TCP connection

→ Browser often open multiple parallel TCP connection to fetch referenced objects in parallel

Example 5 :-

#Q. Consider a web browser at client machine made a request for a web page to a web server using non-persistent HTTP with five parallel TCP connections. The web page contains base HTML file and ten very small image objects. How many total number of RTTs (Round Trip Time) it takes to load the web page completely ?

For Base HTML File → 1 RTT + 1 RTT = 2 RTT

Base HTML File is having 10 small image object reference

i) Use five parallel TCP connections for five objects

1 RTT + 1 RTT = 2 RTT

ii) Create five parallel TCP connection again for remain five object

1 RTT + 1 RTT = 2 RTT

Ans = 6 RTT

Topic : Persistent HTTP (HTTP/1.1)

→ Server leaves connection open after sending response

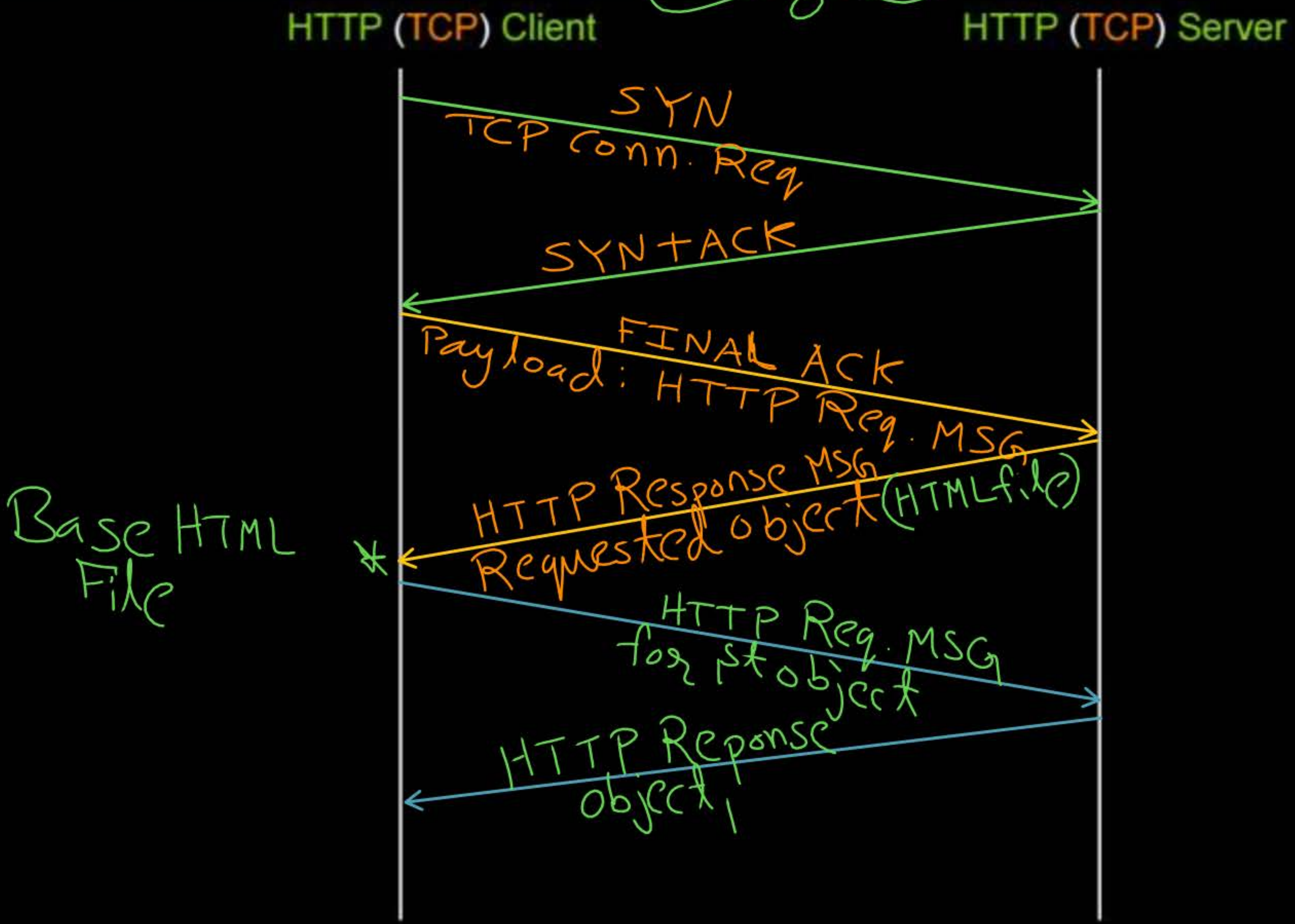
→ Subsequent HTTP messages between same client/server sent over open connection

→ Multiple objects can be sent over single TCP connection [between Client and Server] [if all the objects are placed in same Server]

same

⇒ client have to initiate close connection at the end.

Topic : Persistent HTTP (HTTP/1.1)



Example 6 :-

#Q. Consider a web browser at client machine made a request for a web page to a web server using persistent HTTP. The web page contains base HTML file and ten very small image objects. Assume base HTML file and all ten image objects are placed in same web server. How many TCP connections required to load the web page completely ?

Ans: ONE TCP Connection

Example 7 :-

#Q. Consider a web browser at client machine made a request for a web page to a web server using (persistent HTTP) The web page contains base HTML file and ten very small image objects. Assume base HTML file and all ten image objects are placed in same web server. How many total number of RTTs (Round Trip Time) it takes to load the web page completely ?

(1) ONE RTT → For TCP connection
(2) ONE RTT → For Base HTML File

Base HTML file contains 10 referenced URL for 10 small image objects

i) ONE RTT → for 1st image object
ii) ONE RTT → for 2nd image object
⋮
x) ONE RTT → for 10th image object

$\boxed{\text{Ans} = 12\ \text{RTT}}$

#Q. A graphical HTML browser resident at a network client machine Q accesses a static HTML webpage from a HTTP server S. The static HTML page has exactly one static embedded image which is also at S. Assuming no caching, which one of the following is correct about the HTML webpage loading (including the embedded image)?

IIT-KGP (H.W.) [GATE-2014, Set-2, 2-Mark]

(A) Q needs to send at least 2 HTTP requests to S, each necessarily in a separate TCP connection to server S

(B) Q needs to send at least 2 HTTP requests to S, but a single TCP connection to server S is sufficient

(C) A single HTTP request from Q to S is sufficient, and a single TCP connection between Q and S is necessary for this

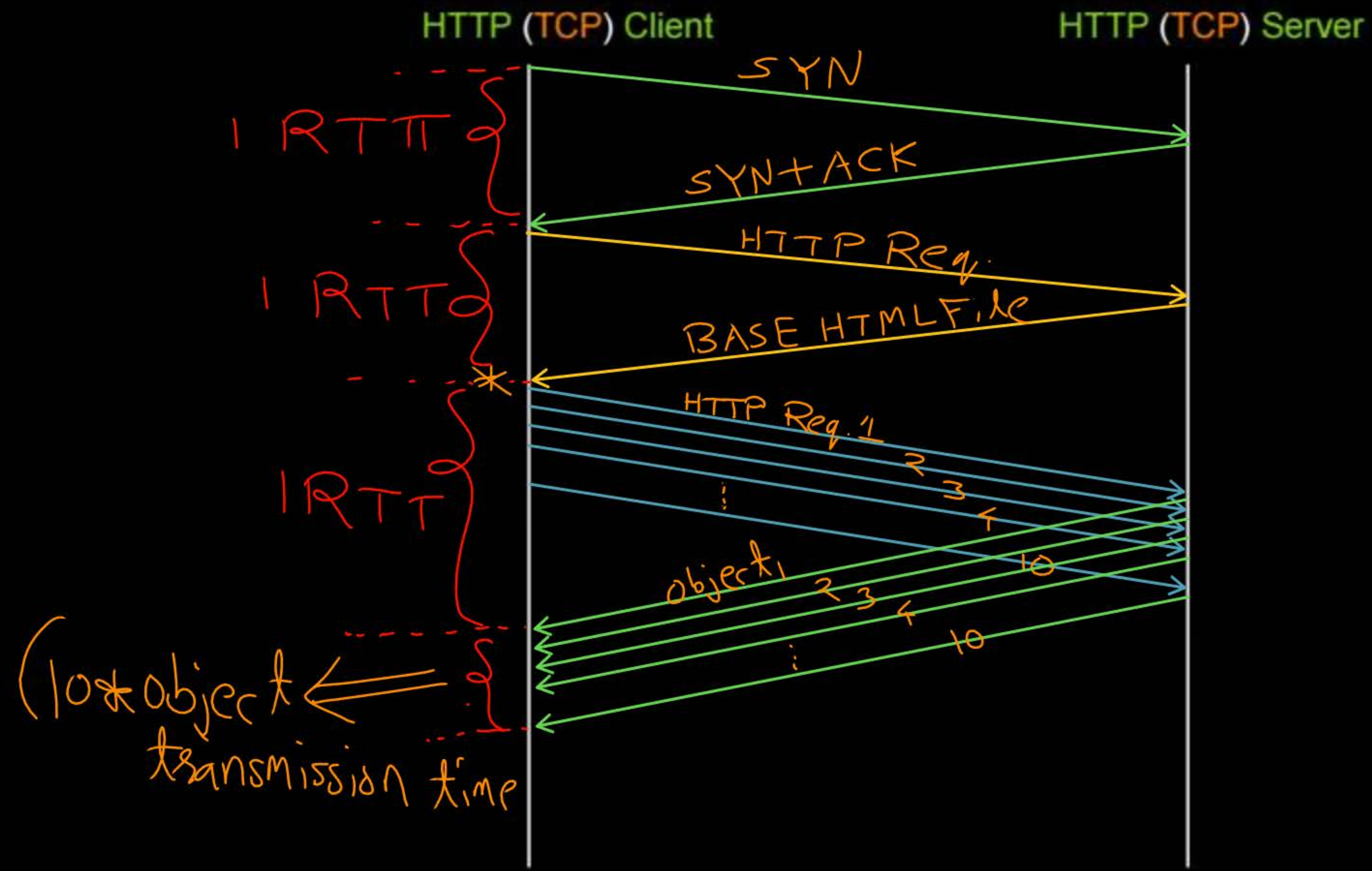
(D) A single HTTP request from Q to S is sufficient, and this is possible without any TCP connection between Q and S

Topic : Persistent HTTP (HTTP/1.1)

→ Client sends requests as soon as it encounters a referenced object

→ One RTT for each referenced objects (instead of two)

Topic : Persistent HTTP (HTTP/1.1)



Topic : Persistent HTTP (HTTP/1.1)

→ Client can make requests in pipeline

→ Server can send referenced objects in pipeline

→ One RTT for all referenced objects (for small size objects)

Example 8 :-

#Q. Consider a web browser at client machine made a request for a web page to a web server using persistent HTTP with pipelining The web page contains base HTML file and ten very small image objects. Assume base HTML file and all ten image objects are placed in same web server. How many total number of RTTs (Round Trip Time) it takes to load the web page completely ?

1 RTT → TCP connection
1 RTT → for Base HTML file
1 RTT → for all 10 objects in pipeline

3 RTT

$\boxed{\text{Ans} = 3}$

[MSQ] IIT-K, H.W. [GATE-2023][2 Mark]

#Q. Suppose in a web browser, you click on the www.gate-2023.in URL. The browser cache is empty. The IP address for this URL is not cached in your local host, so a DNS lookup is triggered (by the local DNS server deployed on your local host) over the 3-tier DNS hierarchy in an iterative mode. No resource records are cached anywhere across all DNS servers.

Let RTT denote the round trip time between your local host and DNS servers in the DNS hierarchy. The round trip time between the local host and the web server hosting www.gate-2023.in is also equal to RTT. The HTML file associated with the URL is small enough to have negligible transmission time and negligible rendering time by your web browser, which references 10 equally small objects on the same web server.

Which of the following statements is/are CORRECT about the minimum elapsed time between clicking on the URL and your browser fully rendering it?

(A) 7 RTTs, in case of non-persistent HTTP with 5 parallel TCP connections.

(B) 5 RTTs, in case of persistent HTTP with pipelining.

(C) 9 RTTs, in case of non-persistent HTTP with 5 parallel TCP connections.

(D) 6 RTTs, in case of persistent HTTP with pipelining.

#Q. Identify the correct sequence in which the following packets are transmitted on the network by a host when a browser requests a webpage from a remote server, assuming that the host has just been restarted.

[GATE-2016, Set-2, 1-Mark]

IISC, H W.

(A) HTTP GET request, DNS query, TCP SYN

(B) DNS query, HTTP GET request, TCP SYN

(C) DNS query, TCP SYN, HTTP GET request

(D) TCP SYN, DNS query, HTTP GET request

#Q. A user starts browsing a webpage hosted at a remote server. The browser opens a single TCP connection to fetch the entire webpage from the server. The webpage consists of a top-level index page with multiple embedded image objects. Assume that all caches (e.g., DNS cache, browser cache) are all initially empty. The following packets leave the user's computer in some order.

(i) HTTP GET request for the index page

(ii) DNS request to resolve the web server's name to its IP address

(iii) HTTP GET request for an image object

(iv) TCP SYN to open a connection to the web server

Which one of the following is the CORRECT chronological order (earliest in time to latest) of the packets leaving the computer ?

IISC, H.W.

[GATE-2024, Set-1, 1-Mark]

(A) (iv), (ii), (iii), (i)

(B) (ii), (iv), (iii), (i)

(C) (ii), (iv), (i), (iii)

(D) (iv), (ii), (i), (iii)

2 mins Summary

Topic HTTP

THANK - YOU

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